

ActInf Textbook Group ~ Cohort 6 ~ Session 6 (Chapter 3, Part 2) ~ 3/11/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 6 (Chapter 3, Part 2) ~ 3/11/2024 | 55:07

all right welcome back

um

we're in the second discussion on
chapter 3 so does anyone just have any
quote or question or anything to kind of
begin
with

any thoughts or we can just start to
look at the the written questions from
before or some page or figure from
three

a a basic sort of disorienting thing uh
I guess in the uh diagram at the top of
page

44 um where it shows external States
internal States action States and
sensory States I'm confused uh about
what are realistic examples of like
influence on that direct Arrow between
active States and sensory States like
again I understand active State or the
point of active State generally is to
impact upon the world and like
eventually we see the effects of that in
sensory States but I guess the direct
Double Arrow between the active States
and sensory States seems confusing to
me yeah it doesn't have to have any
influence in the real
system but it's it it it could but I
think in a if we if the sensors and
actuators of the system are
separate then that edge may have no
effect but in principle it
exists because the the the states are

not like pre-tagged

it's always just based upon the choice

the internal

state so a causal edge here

in sketching out the specific system

You' quickly determine like whether it's

something causal in the system or not

but we'll see if it comes up in any of

the

examples

Susan uh yes um so can you I'm really

unclear in what the what are the

boundaries

between

um the um inference the the external and

internal States in other words

specifically what is action in this

context and I I'm wrestling with speech

acts um you know not doing something is

still an action on your environment if

you think about it

so yes well one aspect of action that is

um not really looked at in the textbook

is deciding when to act like in the kind

of simplest format of the action

perception Loop there's just a

perception and an action every single

time step so the like the choice to like

wait versus to do something at

all um sometimes that's brought in as

like a null action

okay yeah just like an action that kind

of like keeps you where you

are okay there could be a nested model

where the top is a binary decision you

know make a move or not within make a

move there's just the four moves um okay

is is that have anything to do with the

um transition

function the um the the

transition function

is describing how the state the hidden

States change through time so it's the B

Matrix we're getting an

observations okay and then the a kind of

maps the observations to yeah the hidden

States so from the thermometer readings

to the temperature in the room and then

the transition function is the

temperature of the room changing per

ΔT just per time

interval the temperature in the

room underlying

changing and the formulas on the right

there you know that's actually for the

transition

functions yeah so then what okay three

here yep which is B it's saying it's the

distribution of the state at the next

time point okay conditioned upon the

state at the current time point and the

policy so say okay the temperature of

the room at the next time point is

conditioned upon the current temperature

in the room and whether I'm turning on

the air conditioner or not okay okay and

then which policy is like selecting

which slice of b gets applied as the

transition function and one last

question um so is this is this model

just saying that there was a move made

and there was a and that's a SE is that

is

that you know as I'm seeing two

different is that perception action or

is that just a different action

repeating in the picture you know just
looking at the yeah like the fact that
there's two threes
here no actually because you got the
upper and the lower so is that the
higher high road low road no um so 4.3
top graph is the discrete time model and
the lower graph is the continuous time
model okay and then in
um here's just the discrete time model
okay just the top half of figure
4.3 okay so here again we see the B and
it's hidden State at the next time point
to plus one y conditioned upon the
current state and the
policy
okay thank
you and and I think as we kind of may
have alluded to earlier there's
like this representation of the
generative model which is the partially
observable markov decision process and
then there's this particular
partition
representation so here we have the kind
of
hinting at a
circularity and the
flows because the dot on top is the
derivative so the flows across these
four
states which we can associate with the
um from the perspective of these
internal States as the sensory action
and
external and then there's this
form which is more like a branching tree
a little

bit because here like the observations
are a dead

end okay I don't exactly remember what
we got to last week but I think we
looked at figure 3.2

with the Scatter

Plots um does anyone want to ask a
question or should we just try to go
through
these

Thomas oh I'm sorry that was a hiccup by
accident I

apologize okay all right let's look
through some of

these um what's the difference between
the terms basium brain hypothesis
predictive processing predictive coding
active inference so here's it's um
discussed in several

places here's one reference

um these terms can be just kind of
broadly used to refer to very
similar uh theories and kinds of models

or if you really get down to Last Mile
they could differ um basian brain

hypothesis hypothesis suggesting it's
something that could be tested and
falsified for a given system like you

could hypothesize that the brain is
doing basian computation or is doing
something that can be modeled with

basian computation you could hypothesize
that and have criteria for why you would
accept that hypothesis or not and then
you might as well um do that and then
find out or not so this is testable

hypothesis describing whether brains
which are just some systems in the world

but a tiny subset of systems do Bayesian computation however they do it or whatever their distributions are unimodal active inference probably a lot of ways to Define it in a broader sense it could just refer to models that have action included within the scope of inference like models that do sense making and action selection or to be more specific we might be talking about active inference models like perception action models under the free energy principle which is kind of chapter three high road and then predictive processing and predictive coding again these are sometimes used interchangeably but um to get technical about it predictive processing is the kinds of architectures that'll get visited in this textbook where there's a multiscale model and um higher levels are making predictions about lower levels and um the entire processing architecture has to do with predictions hence predictive processing predictive coding kind of nuances that a little bit and highlights that the messages that are being passed include predictions and prediction errors so you have prediction process predictive processing systems that that do predictive coding you could also design one that had like one or the other so they're all really

related but they're not exactly exactly
the same

CH if one defines preferred States as
expected

States then one can say that living
organisms must minimize the surprise of
their sensory observation I'm not clear
on the ontology here is it that
preference is the same as expectation or
that formally we can use them in the
same

equation these are lot of partial
related questions

here

[Music]

I'm not I don't remember if we looked at
this one last

week when we're talking about the
variable called preference

C it's a distribution over
observations I think last week we talked
about how expectation can be used in
kind of two senses one is the
expectation is like about an event that
hasn't happened yet and that's a
conversational usage of what we expect
to happen in the future also a given
distribution can have the expectation
calculated of it that's the fancy e and
that's the same thing is saying the the
central tendency or the mean of that
distribution

um in the case of pragmatic value in
expected free energy equation

2.6 yes when we talk about we're talking
about an expectation about
observations but just saying expectation
alone is almost not a complete sentence

because you need to so it's like saying well I'm talking about the mean but the mean of what variable so there's settings where um it's clear from Context what it means and that we're talking about the same variable but you could also Imagine a situation where you're talking about the central Tendencies or the future possibilities of other variables so it's not the terms are simply identical by definition it's that in certain kinds of generative models they're described the same way to highlight that there isn't the proposal of like a reward function but rather there's a direct surprise bounding on expectation um Aaron yeah well okay this isn't exactly a refinement of this question but it's related I I guess like I I have been struggling with how to think about uh I I guess the distinction between priors and preferences in contexts where you in advance of receiving some observation have reason to expect uh and I think the narrow sense um that conditions will be other than you would prefer right so if the weather report this morning says is going to be 115 in the afternoon and I you know my car rigs down and I got before I I open the car door to get out on the side of the road I expect it to be uncomfortable and challenging to my body's need for homeostasis I guess like in maybe it's not an ontological distinction but I am struggling with like the uh uh under what conditions is

reasonable to sub uh to substitute
um a distribution that describes
preferred States into to a context where
where you would be talking about a
distribution that describes your belief
about States prior to getting more
information about them yeah great
question so let's just say we're talking
about temperature so our um C preference
distribution is going to be like
homeostasis just a a a gaussian
distribution around

37c so we're going to talk about that as
our preferences now it's also a prior
that's why these things are like not
complete sentences it's not that there's
one variable just only and always called
prior because any distribution that
could be updated you might be talking
about the prior distribution of that
distribution so hence it's it's like
when talking about like beliefs it's
like about which one of these priors
prior distribution about which one now
when people don't further Define which
prior they're talking about usually
they're talking about D the prior on
hidden States

so let's just say that um we're we're in
the desert and we we have an accurate
thermometer so our a matrix is is just
the identity Matrix just the ones on the
diagonal so we're just getting accurate
thermometer

readings um we prefer 37 in terms of our
homeostasis but our prior belief is that
it's going to be

50 and then we get a thermometer reading

of 50 and then that's consistent with the hidden State estimate of the temperature being 50 so there's totally no issue with preferences over observations interfering with the sense making process because C doesn't come into play with a or D however the the C does shape action through a pragmatic value but C doesn't play a role in sense making or in um estimating future time steps at all is there a minimal example of a thing is it correct to say a thing is the temporary Persistence of a marov blanket perhaps that it's the Persistence of something that can be modeled as or with a Maron blanket are all things then just not an equilibrium steady state processes well considered one way they're at an equilibria point to allow them to be measured but by virtue of our scenario overall being dissipative the fact that they're existing is non-equilibrium do we need to be more strict in saying these things are are only self-evidencing things rather than things in the colloquial sense totally depends on what what's being discussed if it's an inner particle then it may not be providing any extra information to say that it's

self-evidencing like to say that rock
is fulfilling predictions that it
expects to be there
um first issue you may not be adding any
richness to the account and second off
you're proposing a map that's that's
indicating some causal
um capacities for the rock that are
unrealistic whereas when you start
getting into that squirrel is acting
like
XYZ then you are creating parsimonious
accounts of behavior that can't be
accounted with simpler and you are
making maps that have features that that
system might actually
have so the squirrel self-
evidencing
um it's probably more of a value ad in
terms of the account and it's probably
more plausible
mechanistically what is the counterpart
in an active agent to conserve versus
disputed physical
systems what is the analogy to friction
in a cognitive system how strong is this
analogy that's an interesting question
to explore one possibility is like what
is heat and what is
work in basian
mechanics like if if all of the
information of the observation were
used for belief updating that would be
like extracting 100% of the efficiency
of the fuel if you if you perfectly
burnt the
fuel the maximum extractable heat as
work would be like

from the hydrocarbon down to whatever the um terminal product of burning was like you could burn the candle and extract all the heat from the candle um or like how they test how much calor are in food like you just burn it and there's a Max am of energy to extract so then by analogy there's like a maximum amount of information to extract from a given signal like described by its entropy so there might be something like how much of the entropy of the signal is going into belief updating versus how much is being um partitioned to like dis

itive but I'm not sure about friction specifically because friction isn't just the same as

heat okay chapter 3 High

Road starts from premise that to survive any living organism has to maintain itself in a suitable set of preferred States these preferred states are first and foremost defined by Niche specific evolutionary

adaptations however as we will see later in advanced organisms these can also extend to learned cognitive

goals I'm curious about the

applicability of active inference to

social structures such as economic

systems firms self-organizing groups and

government systems amongst others one

can argue that social systems have

preferred States created from both

evolutionary Niche and learn cognitive

mechanisms has active inference yet been

applied to these sorts of

systems

yeah cool that some people have added

some

links yes applied to a limited

extent AI explains in the intro video

Markov blanket is a boundary in the

state space not necessarily a spatial

temporal boundary

if we were talking about a spatial

temporal State space like GPS

coordinates comma time then a boundary

in the GPS comma time

space would be a state space boundary

corresponding to a spatial temporal

boundary but if our state space were

talking about something that wasn't

spatial temporal then the mark called

blanket is not necessarily spatial

temporal even in the map even before get

getting into the map territory stuff

what allows the agent to be

statistically separated from its

boundary as observed there's

blanket so I take it the blanket is a

lens through which the agent observes

their

environment

yes

yes would this lens be defined by their

affordances yes just like on the inbound

it's kind of like glasses like they're a

real like a lens and then like light

comes in and gets focused in but then

also if it were a light source it would

defract out the other way it's not

exactly the same because inbound

perception isn't exactly the same as

outbound action but it is like a two-way

lens that has communication across
interface

Susan wait unmute but then yes

Susan oh sorry um

so I was listening to one of the uh
videos and I've heard um priston say
that Markoff blankets were you were
really just about computational
models what do you have to say about
that like he meant that they were being
introduced as a computational tool
rather than like an ontological Claim
about reality or what did you okay okay
that makes sense thank

you like let's just say we had um 10
variables we could test a model that
every variable influenced each other so
um and and and every three-way
combination matters uniquely and every
five way and seven and nine way it's
like we could test kind of the the
fullest possible model but that's more
parameters to
estimate so it reduces the St
statistical power per parameter so this
is a fundamental trade-off in
statistical

modeling which is like um the fewer
variables you include for a given sample
size the more statistical power you
have however you run the risk of
partitioning variants
inappropriately whereas the more
variables you
include that may come closer into
alignment with what is like believed to
be kind of like a a very
multicausal

Nexus so it can kind of feel nice and appropriate to enable the possibility of of intersections mattering for causality however that can make the causal State spaces very large diluting the statistical power to detect anything and so models are chosen to be at a trade-off point between strongly explaining two few factors and weakly explaining too many factors and that's why how many parameters are included in that kind of best performing trade-off model that kind of parameter estimation is about the map not about the territory does the distinction between active learning and active inference have to do these Imports I don't remember exactly which paper the active learning was but it kind I remembered that like again like sometimes the term is used in a narrow or a broad sense like in a broad sense Active Learning and active inference are the same thing they could both be used to describe a system that's active and learning but then if you really get into it a paper might say well we're using this to mean this specifically so without the link I I I don't remember exactly what what was meant by Active Learning are internal State parameters referring to the edge of the marov state space no they're referring to the inside the marov blanket is The Edge

like the crust around the internal State
parameters

is there any identification for the
edge of a Markoff

blanket like

whichever whatever variable you're going
to point to and say that's the one that
I'm going to call

internal so that's just that's a choice
by the model which one is going to be
called

internal then all of the crust around it
all of the envelope that encloses it is
the blanket of that state but if you had
chosen a different state to choose
internal you would have ended up getting
a different set of blanketing

noes it's kind of like let's just say um
you know we're we're doing Google Maps
and so we we're going to say

um for this trip from A to B this is the
first straight this is the last smile
and then this is like the space between
but if you had chosen a different
Journey maybe the street that was in the
in between for the first journey is
going to be the first mile for this
other

journey in advanced organisms preferred
states can also extend to learn
cognitive goals

and in advanced organisms like human
they can achieve preferred states by
increasing the abstract social cultural
strategies my question is if they are
cast in terms of active inference must
all of these preferred States and
strategies be ultimately linked to

survival in some way or in the case of advanced organisms can free energy be related to something other than survival if acting that way was in some way compatible with existence you wouldn't see it if acting that way was deleterious to Fitness you'd see it less so that's the kind of um Evolution like component which is like it doesn't mean that every single phenomena was necessary and sufficient for existential success but certainly measuring something means that it wasn't incompatible with existence so then by the time that we're talking about organisms that might be um with all kinds of different cognitive phenomena especially if they're going to be like in like a um a new or changing Niche you could have um maladaptive you have a ball rolling downhill and you could just be observing the last five minutes of a maladaptive trajectory of the ball rolling downhill so like past returns do not guarantee future returns or whatever they say can markov blankets be considered as decoupling an agent in an environment yes you could think of it as how they are coupled which is to say how they are decoupled like how they are articulated the joint is both how things are connected and how they're separated can surprise only be diminished through action

so here's a quote from chapter 2 um
pointing to the two ways that
discrepancy is addressed in active
inference change the mind or change the
world so if we're in the temperature
situation we prefer it to be homeostatic
and the temperature is higher than we
expect then we can either take
affordances that we believe are going to
bring the temperature back into
alignment with our observation
preferences that's pragmatic value
change world or we could learn and come
to um dissipate the discrepancy just by
updating what we expect things to
be however for like a physiological
prior those may not be
learnable

huh um I

guess uh what happens if observation uh
were surprising and then they fall back
into um non not surprising um values
like the temperatures spiked up
unexpectedly but then it just came back
down to
normal

yes that's a good question so like
um a related question is like what
updating strategies are adaptive under
what patterns of environmental
regularity so if the environment was
like very slowly changing then a very
slow changing
um belief updating might be appropriate
if the environment were very rapidly
changing then you might want to track it
more rapidly if the environment had like
one in 10 times let's just say there was

like just a erroneous measure so if it was like a noisy environment where sometimes you got like wild measurements but then they were just like part of a noise then you um would have something like a CET filter which is basically

What's Happening Here in the temperature example this kind of looks like CO2 measurements in the in the environment too but it's kind of like you have these noisy measurements coming in and then the colan filter is basically saying how can we smooth

this um in the extreme case it Smooths just taking the moving average in the extreme slow case in the extreme fast case the smoothing is just tracing exactly what you're getting but then there's somewhere in between that like Smooths out some of the noise guess could you put this under the umbrella of um the action of updating your own

beliefs yeah like moving a belief distribution could be understood as a cognitive action like if you had the choice to do it or not or if you had the mental action to attend if you had the um you have mental action on the dial of how much to pay attention to it zero attention to it means it's going to have no impact on your beliefs because if you don't pay attention to something it's as if it didn't happen 100% attention to it means you're going to update your beliefs 100% to exactly whatever the the last thing that was said was so then that would open up an internal

possibility of how much should I attend to this environment so then in a noisier environment you would want to reduce your attention to make your smoothing slower to average out more whereas if you in a more precise environment with like more um relevant or Salient information you'd want to increase your attention to to maximize information gained but then when we're measuring supplies and you know the you know the temperature spikes up and then it spikes back down I guess you could say that Spike itself is is a surprise but then now your your prize for the temperature is kind of normalized again so yeah I guess you know like it could be a little bit confusing on how you think about um you know supplies is only minimized through action but in this case it kind of feels like it was minimized just from the itself yeah interesting Susan so is there a a specific specific parameter or computation uh to account for the capacity constraints and like cognitive capacity or cognitive control you know when people slip into a state of overwhelm they're going to change States so yeah there's that would be a back to the transition but I didn't know if it was that that's really interesting I

mean let's sketch it out and explore it
because I think that's we're not going
to find that in the
minimal PDP maybe just be a state change
I guess yeah but if we if we um let's
just say that we uh we imagine two
internal states that the agent can be in
one or leg what about
um so how about instead of um too cold
just right and just hot yeah there
there's an internal model that basically
it's the temp you know is this epistemic
is this too boring just right or is this
too
much that may not
be um like that doesn't need mean that
that's the ultimate truth like you could
have a temperature estimating agent that
actually thinks that there's a rooms
that are too hot but actually it could
do just fine to them just because it's
the agent's estimate doesn't mean that
it is that but so now there's there's um
the agent is doing um estimate it's
getting Challenge on the y-
axis and then it's estimating how
difficult is this
content and then it's doing some
estimate and then like the the accurate
estimator would have um basically would
would have would know when it was in
boring just right or too hard whereas
you could have an optimistic estimator
that you know skews to one way or the
other way and then each of those three
states might be associated with um like
maybe in the in the just right phase
maybe that's when there's a feeling like

there's the highest amount of
intentional control but then if
something's too easy or too hard
somebody might feel like they like
there's no point in exerting cognitive
control because it won't
matter well it'll matter if you have
like ADHD
and so yeah managing the that uh I mean
it's the energy it's
the um
entropy comes back to the
entropy yeah to to ordered or too
complex yep is does not
have yeah that would be good to add to
that graph the flood graph to redundant
or to novel
yeah how's markco blanket a restatement
of the classical action perception
cycle there probably a lot of ways
to
um it's a bit of a
restatement perhaps because
um that's so funny like cybernetic agent
like I was
expecting like
this but hey I mean oh so for for many
decades I mean this
is more similar than not to what we've
just been discussing with predictive
processing and active inference incoming
sensors with a reference prediction and
then the differential goes on to
influence the
actuator that acts upon the external
system which then responds and then
passes another sensor
observation um so

conceptually like no one really
disagrees that observations are coming
in actions are going out and there's
like causation on both sides or you know
so the reason why the marov blanket is a
bit of a restatement is because this
cybernetic Loop was phrased before the
basian causal graph literature of the
1990s like with Judea Pearl
so octave inference
restates the classic action perception
cycle in terms of the causal graphs
that have existed since the
1990s I will say I want to say that uh
that the cybernetic um viable systems
models come a long way too so don't
throw the baby out with the bass
water no definitely not um Dave Douglas
is uh working a lot with American
cybernetic society and um and I think
that combining our Journal work with the
uh International Society for system
science and cybernetics like it will
reveal many
many yeah parallels like
this these are these are like it just
here it's like well the circle's kind of
conceptual but now we have a technical
definition for that boundary in terms of
causal graphs mark blanket or in terms
of category Theory but
um like Robert
rosen's
ecology I don't know how useful the um
Wikipedia is but he developed a lot of
category Theory ecology models again
ones that that pre-s a lot of what
were like

here but it
just it just that it's it's a schematic
so it's very
evocative what's really different now is
there are open source software packages
and first principles
formalism that's like this is
literally
me bar more similar than
not even this looks like those marov
blanket uh Scatter Plots in the
chapter it just so so there's just
there's more similarities than
not
hence trying to get them
together living system
um resolve fundamental biological
Problems by exerting active control over
states is the way to understand why
automatic mechanisms are active in
thinking that these are still involved
in calculating
EF but the time Horizon is small is
there another concept of passive control
or is all control
active take the example of touching the
hot
kettle
probably a lot of ways to think about
this
one
um but the automatic mechanisms being
active like I I don't think that they
always need to have
planning like for example secreting
insulin and
glucagon and hormone regulation of blood
sugar that could just be reflexive

actions like as something gets to here
then more of this hormone is secreted
and then I it goes down more of that
hormone is secreted so that's still
active control but it's just reflexive
control is there a concept of passive
control that's almost like action
without action or like the ability to
like control control something without
acting which you could probably design a
situation where that does happen like
where um by by

non-action the agent has the other
system um aligned with its preferences
but suffice to say it wouldn't be doing
so by virtue of actions that it took um
Erin yeah I guess riing on this and um
in the chapter and a few other places in
discussion um the the concept like
temporal depth of a system and like its
ability to plan has come up and I guess
like this is this is partly in this
question we're talking about like some
actions that a system undertakes are
informed by that temporal depth but some
actions that the same system undertakes
are really shallow like removing your
hand from the hot Kettle you don't need
to see very many steps in the future to
understand that's like the highest value
thing you can do um but so I guess that
does that that also implies something
about they being like like

a A

heterogeneity in the the thickness of
consideration of actions that the agent
is is

driving yeah definitely like the

organism is a multiscale phenomena so even in the same moment like in might the best map of what I was doing 1 hour ago might have been um regulating uh oxygen homeostasis over seconds time scale but then you know food over a dayong time scale but then like Financial you know well-being over a year so you could make multiple time scales of action even a slow action is happening in a moment and then and then the kind of no planning model first off it's the computationally simplest so it's a great starting point and then you can look for deviations um and then ask where do we need additional phenomena to account for what we're seeing like like birds flocking like the famous models don't involve any planning it just involves the visual field and basically it's like if they're too close like you pull back and then if they're like too far you move towards them or something like that so then if you do a model you find that you can replicate the statistics of this this empirical flock there's no need to invoke planning but then maybe there's a residual with some real system it's like oh actually you know let's try let's see if planning could explain that variance Gap

Susan wait unmute then

yeah um oh

gosh was looking at one of your slides and and

uh so I I I I totally lost in my train
of thought but I will say that I saw a
video back on the the Swarm and the
studies show that a large percentage of
the of the of the Swarm is just
following you know whoever is closest to
them so you can't overthink this
thing yeah that's that's this is a very
funny in the ass like
I I don't I think it's a little bit
sensationalist to call them lazy but it
was it's like most nestmates at most
moments are just standing there that's
excess capacity or could it you know
when if nestmates die or some if there's
like an
alarm and so it's like it if flocking is
the goal
then yeah well I think it I think
there's a link back to the cognitive
capacity um and level of
abstraction in other words if I'm if I
don't understand more than two Dimension
two
dimensions then you I can only
reconfigure two
Dimensions maybe ever thought but anyway
just a
thought
H yeah Hector um he he's been sharing
some resources about the development
that happens in a child between like the
ages of four and seven as they're
separating out like beliefs from
knowledge where they'll do the
experiment like a little character sees
somebody put a ball into a box then the
character leaves the room and then the

the ball is moved and the character comes back and then you ask like where does the character think the ball is and then it's kind of like theory of mind like you have to understand that well the ball is really in a different place but they think it's here so then that um that a l it's the holding up of multiple possible beliefs about where that ball

is but if you didn't have the capacity entertain and like to juxtapose the memory of the ball moving with when they left the room and all that then you would collapse down to either the ball being you know to to to incorrect answer to that question but then what's interesting about it is like one day the child will not be able to do it and then like two weeks later the child will be able to do that and then that capacity will never leave them for life just like on a random day when they're five years

old

yeah okay cool well fun chapter 3 interesting High Road um hopefully we got to some good topics uh chapter 4 is is kind of like where the high and low road are G to they're going to merge back together and chapter 4 is going to be the um the formal treatment as they say of the generative models and that's going to include some kind of like background

material on the causal
graphs and then come to like figure 4.3
which is like the Rosetta Stone image
with a discrete time and the continuous
time generative
models so these are two possible
architectures that treat time
differently that have the kind of top
down characteristics that were discussed
on the high
road and have that low road mod
construction
so thank you all see you next time
thanks Danel bye
byebye